Data Analysis of Student Performance

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I. DATASET

1. DESCRIPTION

The Student Performance dataset is used for analysis in this project. This dataset contains information on students from Portuguese secondary schools and includes demographic, social, and academic variables. Because the dataset contains variables that may influence student performance, the goal of this analysis is to examine how certain factors relate to the final grades of students.

2. SOURCE

This dataset was obtained from the UCI Machine Learning Repository; it is also commonly available through other platforms such as Kaggle. It contains real-world data collected from student records.

3. VARIABLES AND DESCRIPTIONS

Variable	**Description**
school	Student’s school (binary: GP or MS)
sex	Student’s gender
age	Student’s age
address	Home address type (urban/rural)
studytime	Weekly study time
failures	Number of past class failures
G1	First period grade
G2	Second period grade
G3	Final grade

4. EXPECTATIONS

It is expected that variables such as study time, past failures, and previous grades will present a strong relationship with final grade (G3). Demographic and social factors are also expected to influence final grade, though likely to a lesser extent.

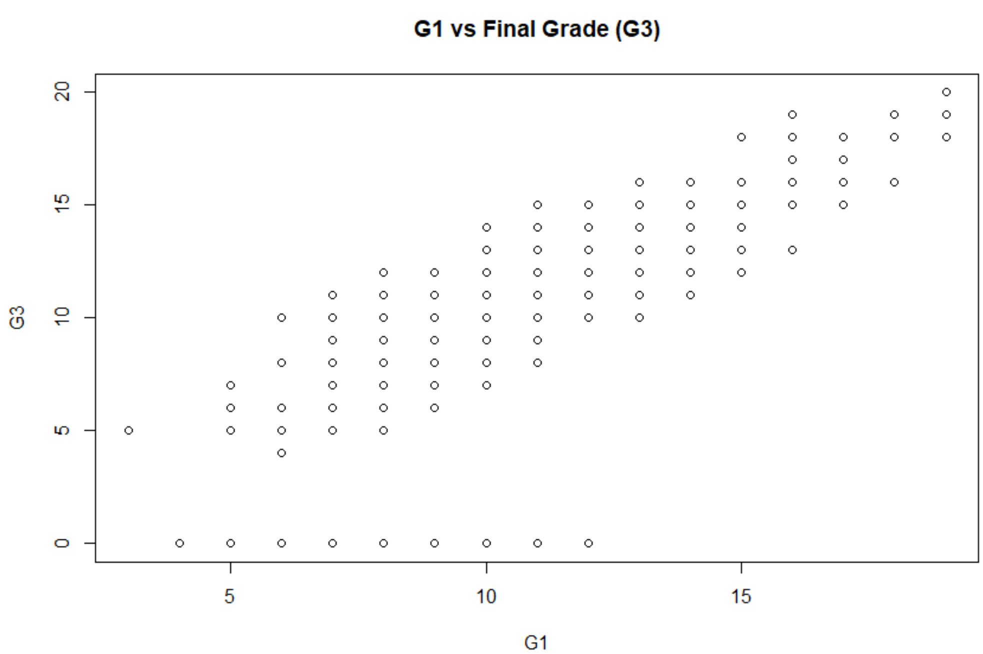
5. PREVIEW

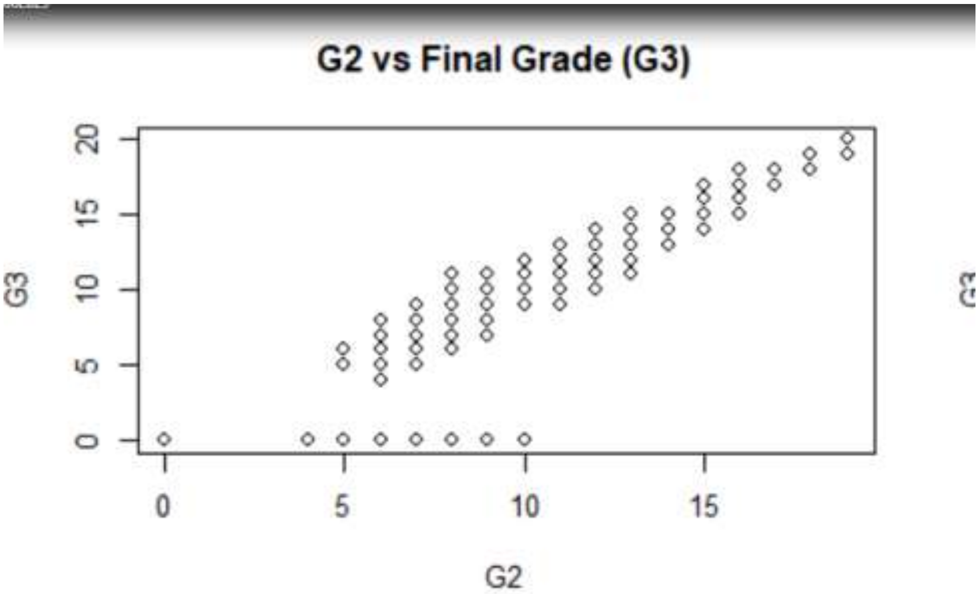
```
> head(data)
  school sex age address famsize Pstatus Medu Fedu     Mjob     Fjob     reason guardian traveltime studytime failures
1     GP   F  18       U     GT3       A    4    4  at_home  teacher     course   mother          2         2        0
2     GP   F  17       U     GT3       T    1    1  at_home    other     course   father          1         2        0
3     GP   F  15       U     LE3       T    1    1  at_home    other      other   mother          1         2        3
4     GP   F  15       U     GT3       T    4    2   health services       home   mother          1         3        0
5     GP   F  16       U     GT3       T    3    3    other    other       home   father          1         2        0
6     GP   M  16       U     LE3       T    4    3 services    other reputation   mother          1         2        0
  schoolsup famsup paid activities nursery higher internet romantic famrel freetime goout Dalc Walc health absences G1
1       yes     no   no         no     yes    yes       no       no      4        3     4    1    1      3        6  5
2        no    yes   no         no      no    yes      yes       no      5        3     3    1    1      3        4  5
3       yes     no  yes         no     yes    yes      yes       no      4        3     2    2    3      3       10  7
4        no    yes  yes        yes     yes    yes      yes      yes      3        2     2    1    1      5        2 15
5        no    yes  yes         no     yes    yes       no       no      4        3     2    1    2      5        4  6
6        no    yes  yes        yes     yes    yes      yes       no      5        4     2    1    2      5       10 15
  G2 G3
1  6  6
2  5  6
3  8 10
4 14 15
5 10 10
6 15 15
```

II. ANALYSIS

2.1 Exploratory Visualization (continuous vs continuous)

```
plot(data$G2, data$G3,
     main="G2 vs Final Grade (G3)",
     xlab="G2", ylab="G3")


cor(data$G2, data$G3)
```

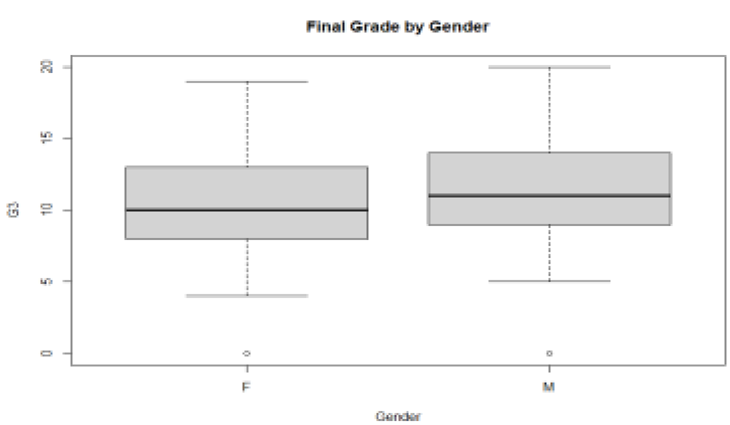




Exploratory analysis was performed to examine relationships between prior grades and final grade (G3). The scatterplots of G1 vs G3 and G2 vs G3 both show strong positive relationships, indicating that students who perform well earlier in the course tend to achieve higher final grades. The correlation coefficients further support this observation, with both G1 and G2 displaying strong positive correlations with G3. Furthermore, G2 appears to have an even stronger relationship with G3. This can suggest that recent performance is the best predictor of final outcomes.

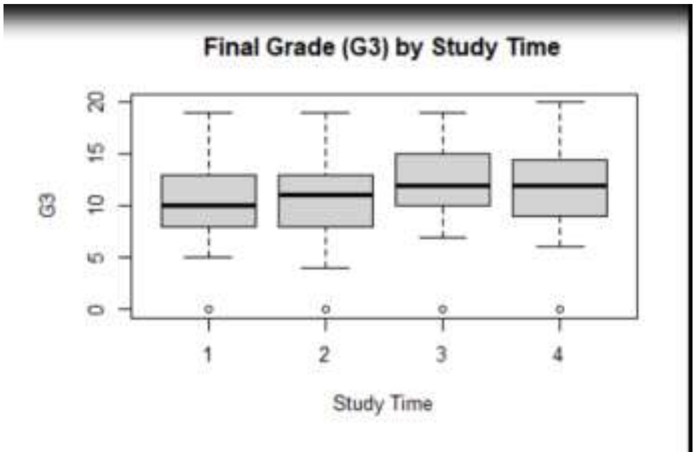
2.2. Categorical vs Continuous (boxplot)

```
boxplot(G3 ~ sex, data = data,
        main = "Final Grade (G3) by Gender",
        xlab = "Gender",
        ylab = "G3")
```



To examine how categorical variables influence final grade (G3), boxplots were used to compare grade distributions across different groups. The boxplot of G3 by gender shows very similar distributions between male and female students, with only a slight difference in median values. This suggests that gender does not have a strong effect on final academic performance.

```
boxplot(G3 ~ studytime, data = data,
        main = "Final Grade (G3) by Study Time",
        xlab = "Study Time",
        ylab = "G3")
```



The boxplot of G3 by study time shows a slight upward trend in median grades as study time increases; however, there is considerable overlap between the groups. This indicates that while increased study time may have some positive influence, it is not a strong distinguishing factor in student performance. These observations are consistent with the ANOVA results, which indicated that study time is not a statistically significant predictor of final grade.

2.3. Statistical Test (ANOVA)

```
> anova_model <- aov(G3 ~ as.factor(studytime), data=data)
> summary(anova_model)
                      Df Sum Sq Mean Sq F value Pr(>F)
as.factor(studytime)   3    108   36.07   1.728  0.161
Residuals            391   8162   20.87
> 
```

A one-way ANOVA was conducted to determine whether study time has a significant effect on final grade (G3). The results indicate that there is no statistically significant difference in mean final grades across study time groups ($p = 0.161$). Since the p-value is greater than 0.05, we fail to reject the null hypothesis. This suggests that differences in study time do not meaningfully impact final grade in this dataset.

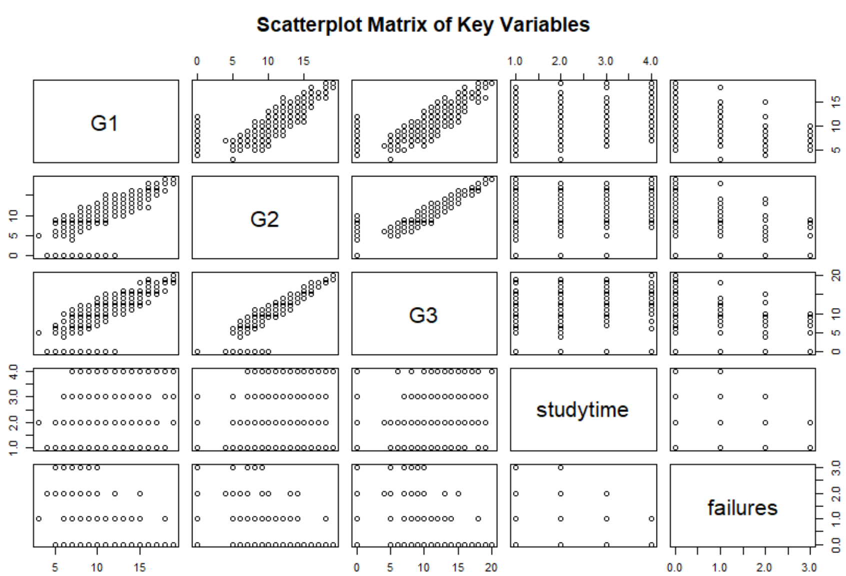
2.4. Correlation Analysis

```
> cor(data$G1, data$G3)
[1] 0.8014679
> cor(data$G2, data$G3)
[1] 0.904868
> cor_matrix <- cor(data[, c("G1","G2","G3","studytime","failures")])
> cor_matrix
                 G1         G2          G3   studytime   failures
G1        1.0000000  0.8521181  0.80146793  0.16061192 -0.3547176
G2        0.8521181  1.0000000  0.90486799  0.13588000 -0.3558956
G3        0.8014679  0.9048680  1.00000000  0.09781969 -0.3604149
studytime 0.1606119  0.1358800  0.09781969  1.00000000 -0.1735630
failures -0.3547176 -0.3558956 -0.36041494 -0.17356303  1.0000000
> pairs(data[, c("G1","G2","G3","studytime","failures")],
+       main="Scatterplot Matrix of Key Variables")
>
```

To examine relationships between key variables, a correlation analysis is performed. The results show a strong positive correlation between G1 and G3 (r = 0.80) and an even stronger positive correlation between G2 and G3 (r = 0.90), indicating that prior academic performance is highly predictive of final grade.

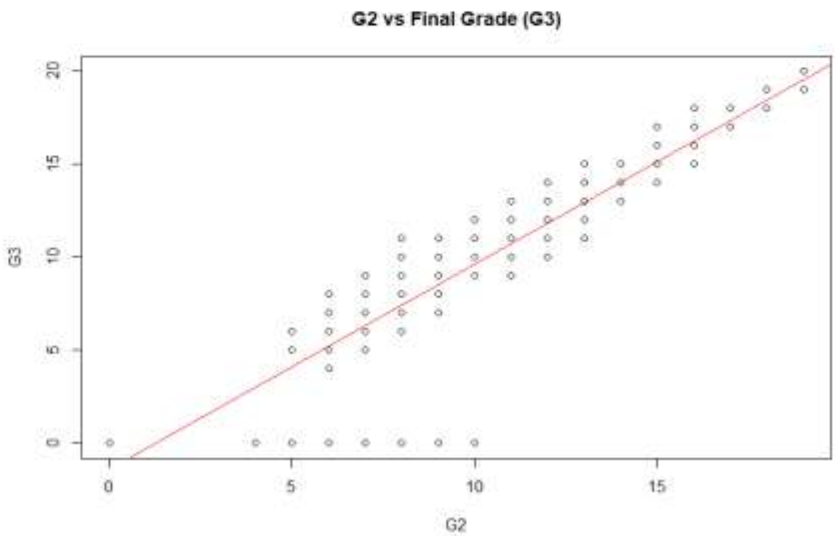
Additionally, G1 and G2 are also strongly correlated with each other. This suggests consistency in student performance over time. On the other hand, study time shows only a weak positive correlation with G3. Failures, however, exhibit a moderate negative correlation indicating that students with more past failures tend to have lower final grades.

```
pairs(data[, c("G1", "G2", "G3", "studytime", "failures")],

      main = "Scatterplot Matrix of Key Variables")
```



To further examine these findings, a scatterplot is conducted. Clear linear relationships are visible between G1, G2, and G3, while weaker and less defined patterns are observed for study time and failures. Overall, these results suggest that prior grades are the strongest predictors of final academic performance.

2.5. Regression Analysis




```
> plot(data$G2, data$G3,
+      main="G2 vs Final Grade (G3)",
+      xlab="G2", ylab="G3")
>
> abline(lm(G3 ~ G2, data=data), col="red")
>
```

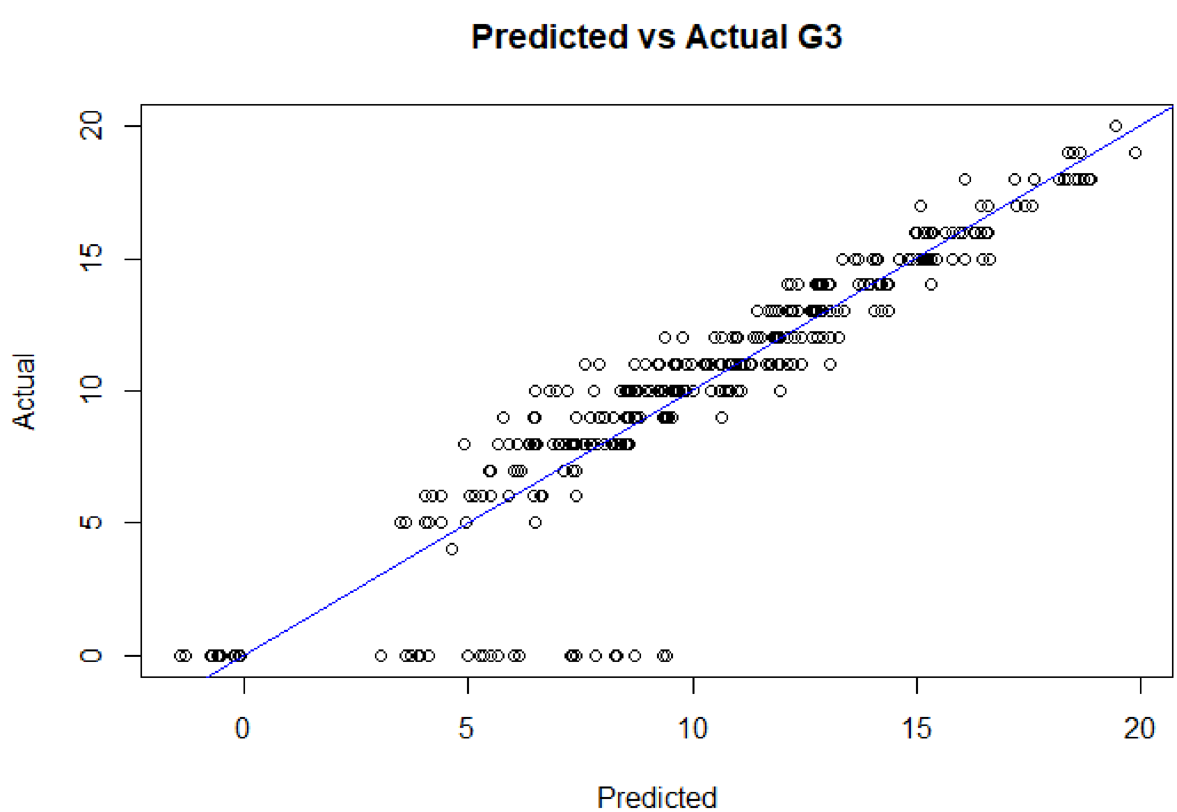
A scatterplot with a fitted regression line was used to examine the relationship between G2 and final grade (G3). The plot shows a strong positive linear relationship, indicating that higher second-period grades are associated with higher final grades.

```
> model <- lm(G3 ~ G1 + G2 + studytime + failures, data=data)
> summary(model)

Call:
lm(formula = G3 ~ G1 + G2 + studytime + failures, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-9.4132 -0.3780  0.2720  0.9855  3.5168 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.19907    0.42944  -2.792  0.00549 ** 
G1           0.15034    0.05639   2.666  0.00799 ** 
G2           0.97653    0.04965  19.666  < 2e-16 ***
studytime   -0.19667    0.11824  -1.663  0.09706 .  
failures    -0.26300    0.14172  -1.856  0.06424 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.928 on 390 degrees of freedom
Multiple R-squared:  0.8246,	Adjusted R-squared:  0.8228 
F-statistic: 458.5 on 4 and 390 DF,  p-value: < 2.2e-16
```

To predict final grade (G3) using G1, G2, study time, and past failures, a multiple linear regression model was developed. The results indicate that both G1 ($p < 0.01$) and G2 ($p < 0.001$) are statistically significant predictors of final grade with positive coefficients.

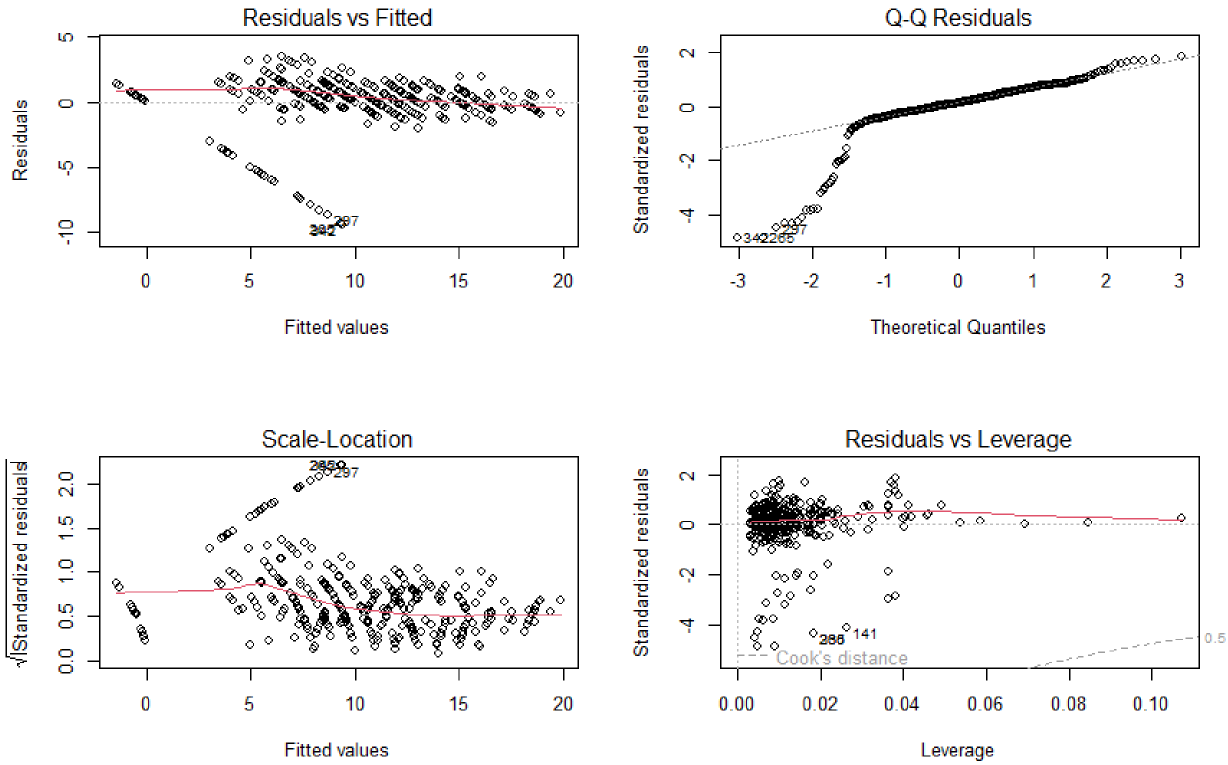
Study time does not appear to be statistically significant ($p \approx 0.097$), which is consistent with earlier ANOVA findings. Failures show a negative relationship with G3 and are marginally significant ($p \approx 0.064$), indicating that students with more past failures tend to perform worse.

The model explains a large portion of the variation in final grade with an R^2 value of approximately 0.84, indicating strong overall predictive performance.



To evaluate model performance, predicted values were compared to actual final grades. The results show that the points lie close to the 45-degree reference line, demonstrating that the model provides accurate predictions. This further supports the effectiveness of prior grades as strong predictors of final academic performance.

2.6. Residual Analysis



Residual diagnostics were implemented to evaluate the assumptions of the linear regression model. The residuals vs fitted plot shows no strong systematic pattern, suggesting that the linearity assumption is reasonable. The Q-Q plot indicates that residuals are approximately normally distributed, with only minor deviations at the extremes.

The scale-location plot implies that the variance of residuals is relatively constant, although slight variation is present. Additionally, the residuals vs leverage plot does not indicate the presence of highly influential observations. Overall, these diagnostic plots suggest that the linear regression model is appropriate for this analysis and provides a reasonable fit to the data.

III. SUMMARY

This analysis examined the relationships between various factors that might affect students' final grades (G3). These factors consisted of academic, demographic, and behavioral variables. The results indicate that prior academic performance is the strongest predictor of final grade (G1 and G2). The exploratory visualizations as well as the correlation analysis showed strong positive relationships between variables and G3. Furthermore, the regression model further confirmed their statistical significance.

On the other hand, study time was expected to have a strong influence on final performance. The ANOVA and regression results, however, indicated that it is not a statistically significant predictor. Demographic factors such as gender were also shown to have minimal impact on students' final grades. The negative relationship between past failures and G3 suggests that students with more prior failures are more likely to perform worse; however, this effect was marginally significant.

To conclude, certain findings align with initial expectations while others did not. Academic variables are certainly strong predictors of performance while study time did not have the expected level of influence. The regression model demonstrated strong predictive capability, as supported by a high R^2 value and accurate predicted vs. actual results. Residual diagnostics further indicated that the model assumptions were reasonably satisfied, suggesting that the model provides a reliable representation of the data.